

Cybersecurity

Machine Learning & Artificial Intelligence

Lesson 2.1



Learning Objectives

By the end of today's class, you will understand the following concepts:

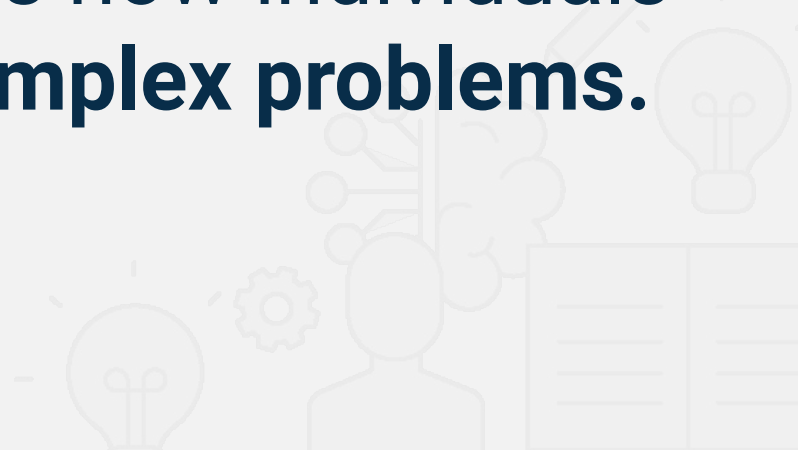
- 1 Definition of AI and ML
- 2 Differences between AI and traditional programming
- 3 Types of AI (narrow AI vs. general AI)
- 4 Importance of AI and ML in cybersecurity
- 5 Ethics and bias in AI
- 6 ChatGPT and similar tools in cybersecurity



Artificial Intelligence And **Machine Learning**

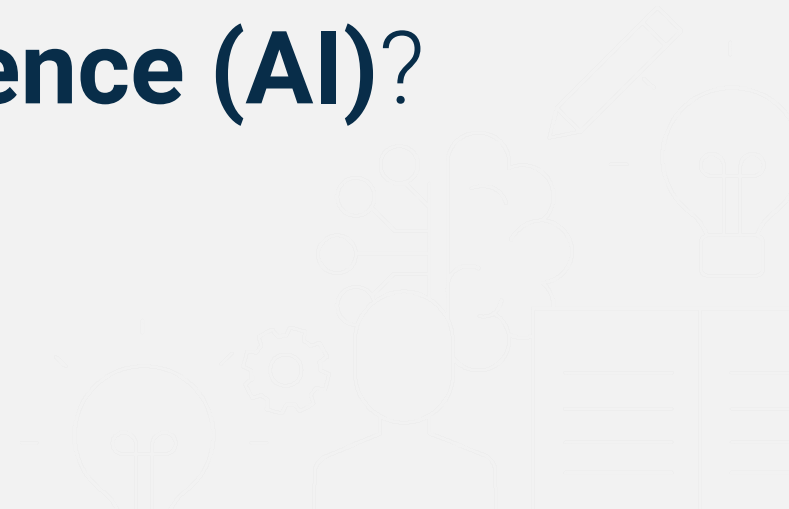


Logic Theorist,
developed in the 1950s,
simulates how individuals
solve complex problems.





What is **artificial
intelligence (AI)**?





Artificial Intelligence is the simulation of human intelligence processes by machines, particularly computer systems.



Artificial Intelligence

It enables machines to perform tasks that typically require human intelligence, such as learning, reasoning, and problem-solving.

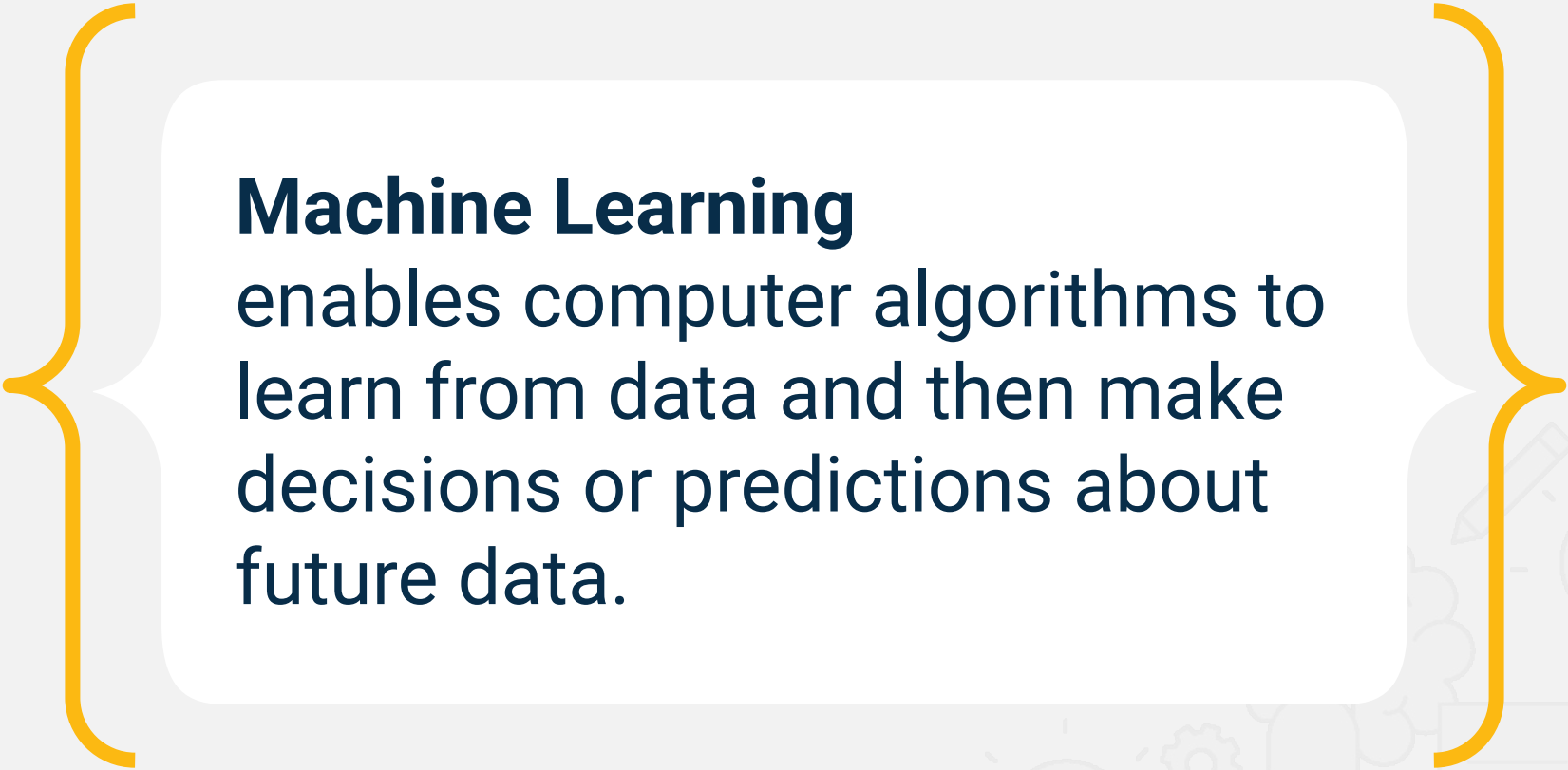
Common applications:

- **Voice assistants:** Siri and Alexa use AI to understand and respond to voice commands.
- **Self-driving cars:** These vehicles use AI to navigate and make driving decisions autonomously.
- **Recommendation systems:** AI powers the algorithms that suggest shows on Netflix or products on Amazon based on your preferences.





What is **machine learning (ML)**?

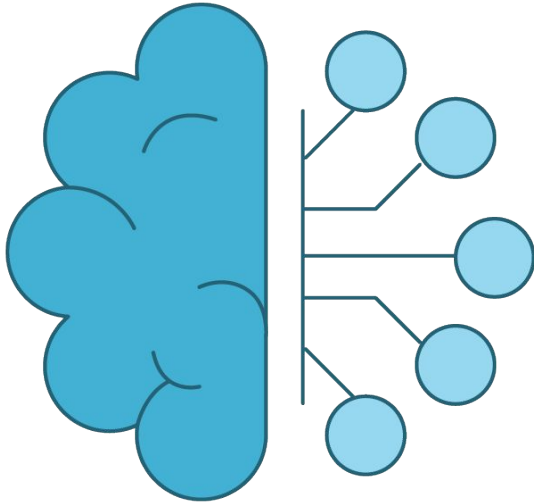


Machine Learning

enables computer algorithms to learn from data and then make decisions or predictions about future data.

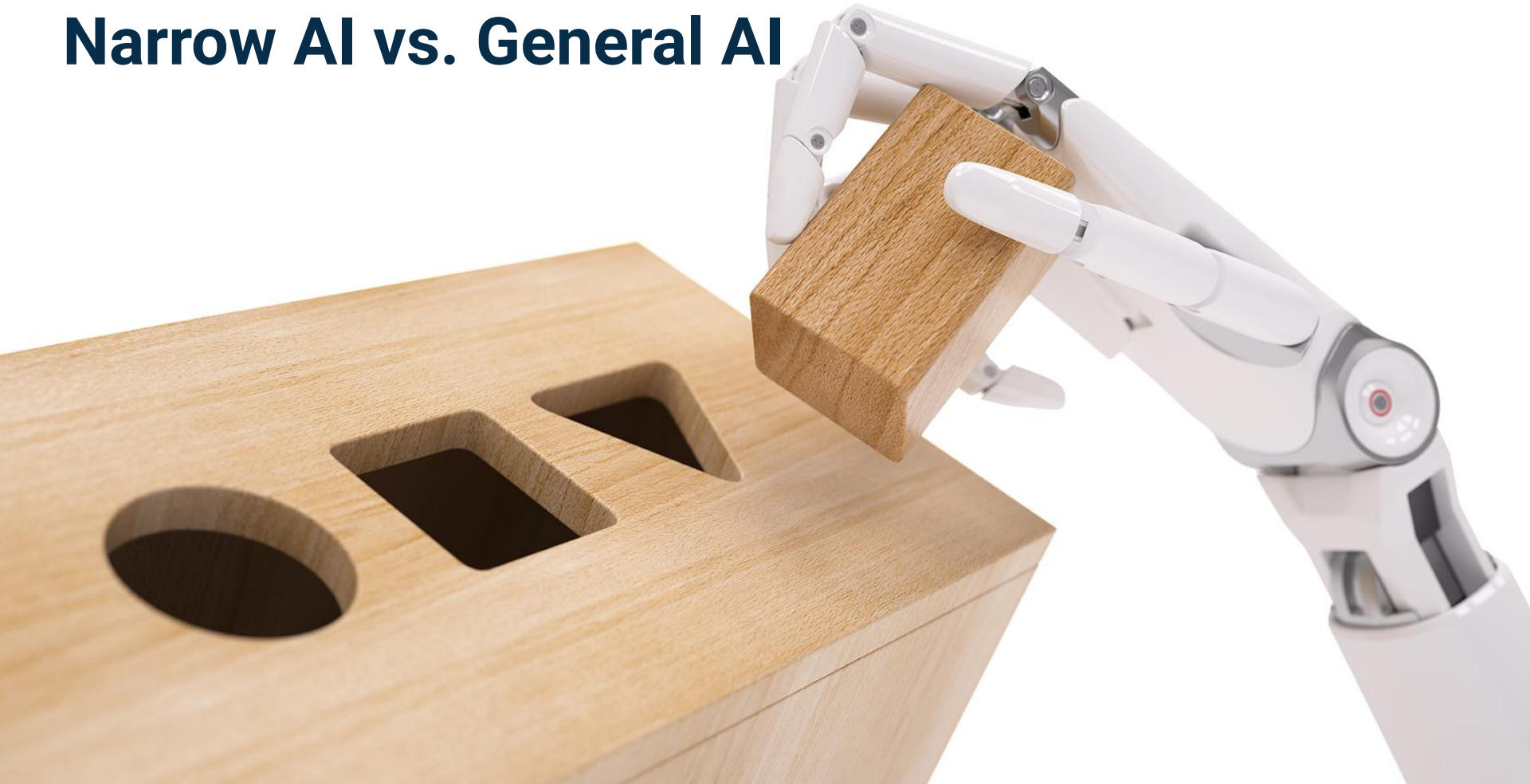


Machine Learning



- Produces predictive models and makes predictions
- Develops rule-based programs with enough data
- Crafts and trains models based on real-world data
- Improves by identifying and disregarding outliers
- Integrates knowledge to make better predictions or decisions
- Does not rely on programmers to manually instruct the computer on how to improve

Narrow AI vs. General AI



Narrow AI vs. General AI

Narrow AI

- Refers to “intelligent” algorithms designed to perform predetermined tasks
- Performs specific tasks
- Unable to transfer knowledge
- Simulates human consciousness but is not conscious

General AI

- The self-aware, independent entities of science fiction stories
- Performs general tasks with minimal oversight
- Knowledge is transferable
- Has human-like consciousness
- May become possible in the future

Importance of AI and ML in Cybersecurity

01

Analysis of vast amounts of data

02

Threat-detection applications

03

Malware analysis

04

Fraud prevention



Importance of AI and ML in Cybersecurity

05

Enhancement of real-time response

06

Automation of routine tasks

07

Predictive capabilities





Questions?





Bias and Ethics in AI and ML

Research with AI

Plagiarism involves copying and using another person's ideas, words, or work without proper acknowledgment. Avoid plagiarism by using AI as follows:

- Use AI for data analysis.
- Generate original insights.
- Cross-verify information.
- Draft with care.
- Maintain academic integrity.

Bias in AI can occur when AI systems make decisions that **reflect human prejudices**, leading to unfair or discriminatory outcomes.



Bias in AI

Biased training data

Data used to train AI systems may contain inherent biases, which can skew the AI's decision-making process.

Biased algorithms

The algorithms themselves can introduce or amplify bias, affecting the AI's outputs.

Biased human decisions

Human choices in designing and implementing AI can lead to biased systems if not carefully managed.

Real-World Examples

- **Facial recognition systems:** Certain ethnic groups can be misidentified due to biased training data.
- **Hiring systems:** Biased hiring algorithms that favor certain demographics can lead to unequal opportunities.





Activity:

Examining Accuracy and Bias in AI

In this activity, you will be using the internet to research recent examples of AI problems in accuracy and bias in the news.

Suggested Time:

20 Minutes



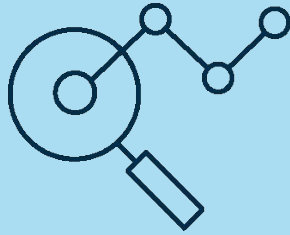


Time's up!
Let's review



Questions?





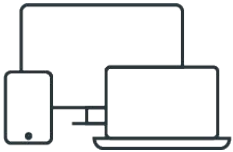
AI and ML in **Data Analysis**

AI and ML in Data Analysis



- **AI and ML revolutionize data analysis.**

- Identification of patterns
- Prediction of outcomes
- Detection of anomalies



- **Data is the foundation of AI and ML.**

- Quality and quantity of data impacts AI performance.
- Algorithms perform better with high-quality data.

Pattern Recognition and Anomaly Detection:

Pattern recognition

- AI and ML models excel at identifying regular patterns within data.
 - Behavior of users in network e.g., login times, access patterns, and commonly used apps.
- AI can detect deviation easily.

Anomaly detection

- AI and ML are adept at detecting deviation from normal patterns.
 - Potential security threats
 - Fraudulent activities
 - Spike in traffic (potential DDoS attack)

AI and ML Applications in Cybersecurity

Some of the applications in cybersecurity we will be discussing in later modules include:

01

Intrusion detection systems



02

Malware detection



03

Fraud detection



Benefits of AI and ML in Data Analysis

01

Speed: Analyzes large volumes quickly with timely insights

02

Accuracy: Improves threat detection accuracy through data analysis

03

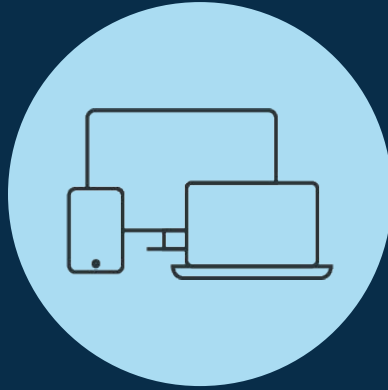
Scalability: Handles increasing amounts of data

04

Adaptability: Learns and adapts to new data and potential threats

05

Proactive threat hunting: Analyzes patterns and predicts security incidents



Instructor Demonstration

Analysis of Logins and Application Access Patterns

Analysis of Logins and Application Access Patterns:

Step 1 – Creating a dataset

Create a dataset of user login times and application access patterns for a network. Include normal behavior (logins between 8am and 6pm, access to specific applications) and a few instances of abnormal behavior (logins at 2am, access to sensitive applications at unusual times). Display the output of this dataset in tabular format highlighting abnormal results.

User Login Times And Application Access Patterns				
	User	LoginTime	Application	IsAbnormal
51	User1	2023-06-13 09:00:00	AppSensitive1	Yes
52	User2	2023-06-15 03:00:00	AppSensitive2	Yes
53	User3	2023-06-03 06:00:00	AppA	Yes
54	User4	2023-06-10 02:00:00	AppB	Yes

Analysis of Logins and Application Access Patterns:

Step 2 – Identifying regular patterns and deviations

Analyze the provided dataset to identify regular user behavior patterns. Highlight any instances where user behavior deviates from these patterns, such as logins at unusual times or access to sensitive applications. Present results in tabular format.

Deviations from Regular User Behavior

This table shows instances where user behavior deviates from regular patterns, such as logins at unusual times or access to sensitive applications:

User	LoginTime	Application	IsAbnormal
User1	2023-06-04 07:00:00	AppA	Yes
User1	2023-06-02 04:00:00	AppC	Yes
User1	2023-06-10 06:00:00	AppC	Yes
User1	2023-06-16 20:00:00	AppD	Yes
User1	2023-06-28 23:00:00	AppC	Yes
User1	2023-06-28 21:00:00	AppA	Yes
User2	2023-06-12 20:00:00	AppA	Yes
User2	2023-06-02 05:00:00	AppA	Yes
User2	2023-06-21 03:00:00	AppD	Yes
User3	2023-06-09 05:00:00	AppB	Yes
User3	2023-06-29 22:00:00	AppB	Yes
User3	2023-06-24 22:00:00	AppD	Yes
User3	2023-06-28 06:00:00	AppA	Yes
User3	2023-06-08 06:00:00	AppB	Yes

Analysis of Logins and Application Access Patterns:

Step 3 – Providing a summary of the deviations

Summarize the deviations in tabular format:

-  Speed and efficiency
-  Accuracy
-  Consistency
-  Scalability
-  Real-time analysis



Activity:

Exploring AI and ML in Anomaly Detection

In this activity, you will use AI to research how AI can detect anomalies using AI.

Suggested Time:
20 Minutes





Time's up!
Let's review



Questions?





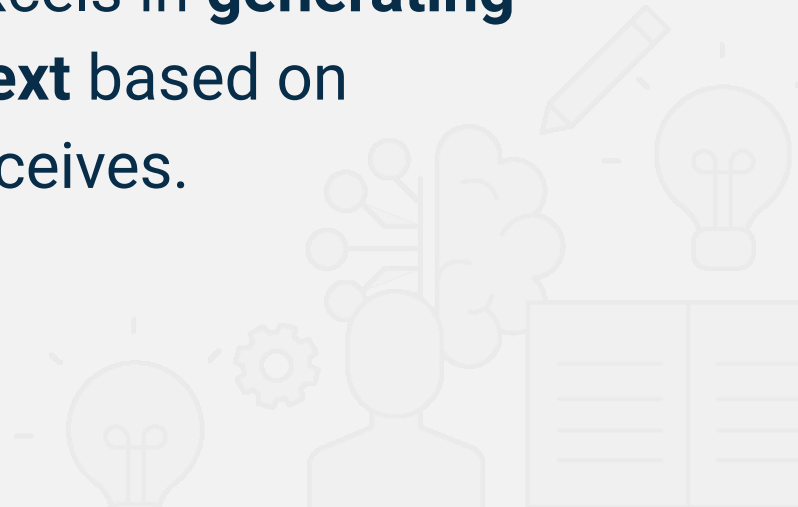
Break
15 mins



Practical GPT



ChatGPT, developed by OpenAI, is a type of artificial intelligence model that excels in **generating human-like text** based on the input it receives.



Key Aspects of GPT

Generative: GPT can create new content. E.g., it can write essays, generate code, compose poetry, and even answer questions.

01

Pre-trained: Trained on a large dataset with sources from the internet. Training helps it understand language patterns, grammar, facts, and even some reasoning abilities.

02

Transformer: Architecture used in GPT is effective for processing sequences of data, like text.

03

Applications: GPT can be used for a wide range of applications, including chatbots, language translation, content creation, summarization, and more.

04

Examples of How ChatGPT Can Help Cybersecurity

01

Automated threat intelligence

02

Anomaly-detection assistance

03

Incident report generation

04

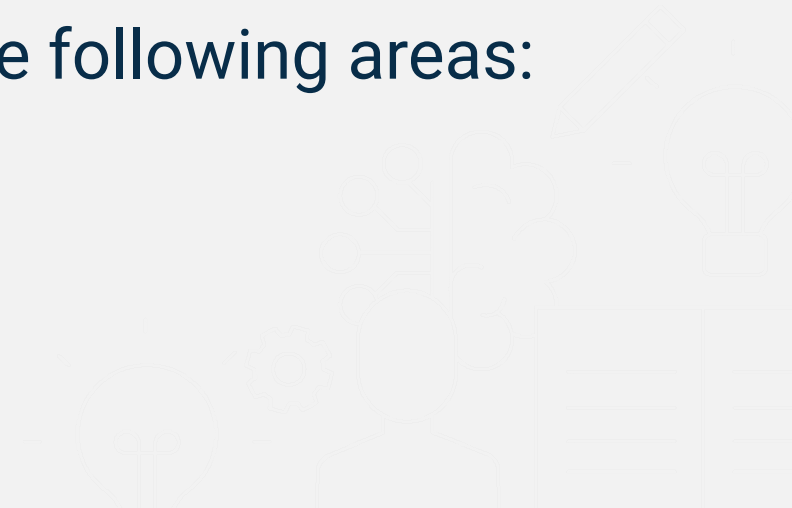
On-demand training

05

Knowledge base maintenance

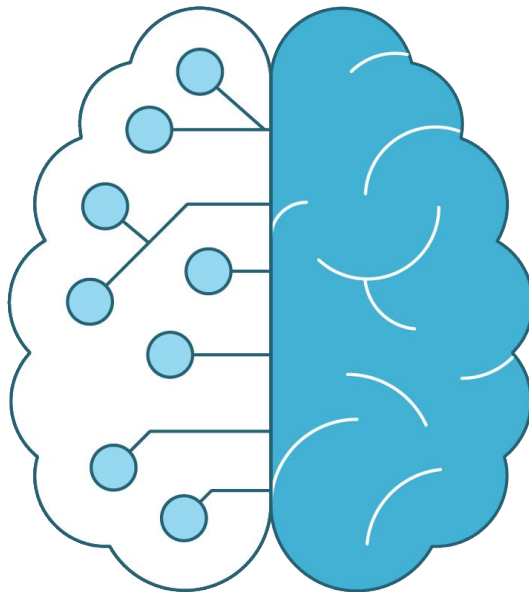


ChatGPT is a tool used by cybersecurity professionals and can assist in the following areas:



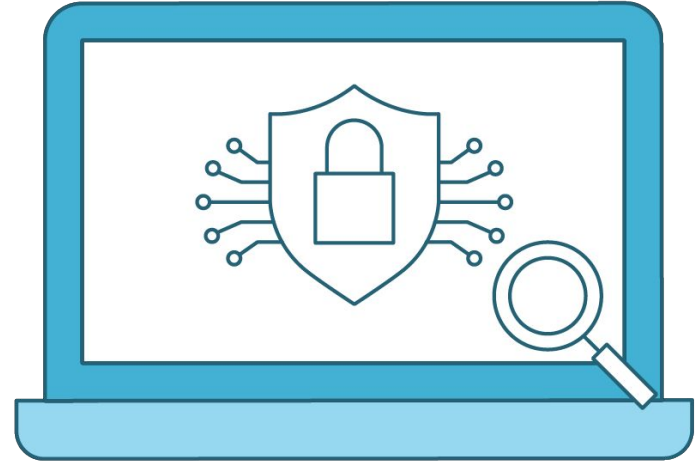
Automated Threat Intelligence

- Processes vast amounts of data
- Summarizes threat intelligence reports
- Stays updated with the latest threats
- Extracts key information from threat reports
- Presents summaries
- Highlights critical details and trends



Anomaly-Detection Assistance

- Assists in identifying unusual patterns and anomalies
- Analyzes user behavior that might indicate a security breach
- Saves time and improves accuracy

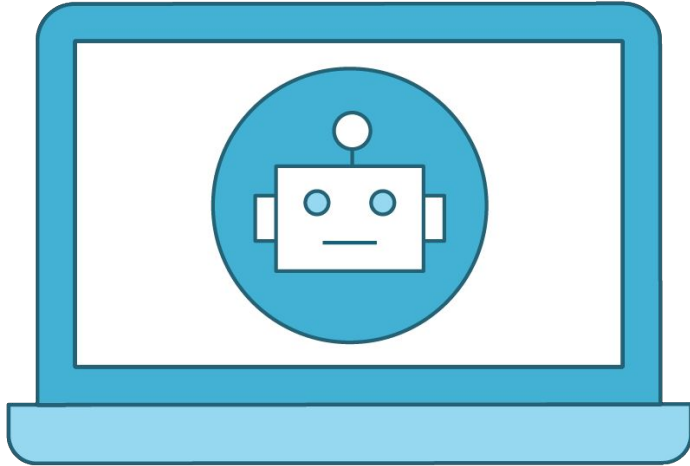


Incident Report Generation



- Helps draft detailed incident reports
- Organizes and summarizes event details, steps taken, and outcomes
- Generates structured report

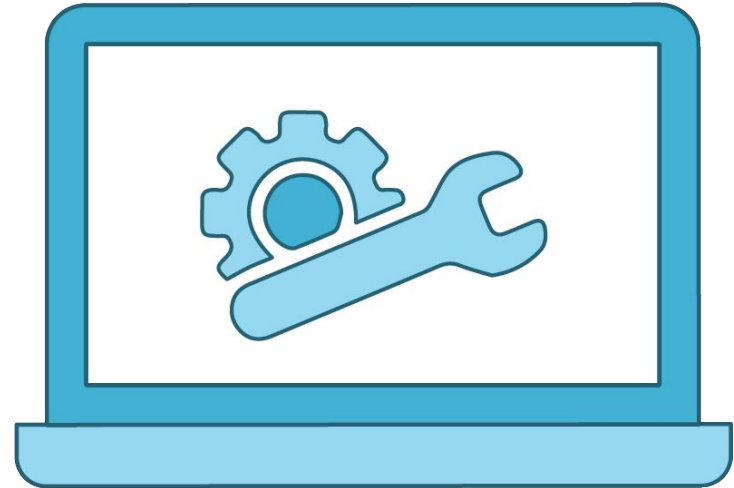
On-Demand Training

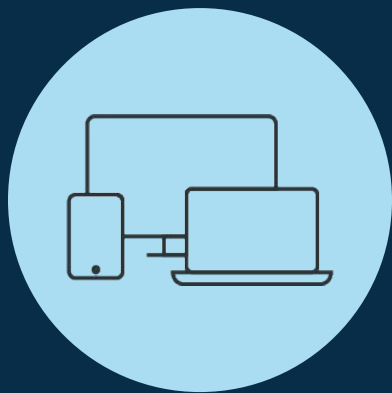


- Provides explanations and answers to complex security concepts and procedures
- Instructs new personnel on:
 - Cybersecurity best practices;
 - Protocols; and
 - Terminologies

Knowledge Base Maintenance

- Helps maintain and update an organization's knowledge base
- Ensures access to the latest information and resources through regular updates





Instructor Demonstration

ChatGPT Demo



Activity:

ChatGPT in Cybersecurity

In this activity, you will explore various prompts within ChatGPT and discuss the responses within your groups.

Suggested Time:

20 Minutes



Time's up!
Let's review



Questions?





Let's recap



Recap

Artificial Intelligence (AI) and Machine Learning (ML)

01

Differences between AI and traditional programming

02

Narrow vs. general AI

03

Importance of AI and ML in cybersecurity

04

Ethics and bias in AI

05

ChatGPT and similar tools in the cybersecurity industry



Questions?





Thank you